Project: Monte Carlo simulation of two dimensional systems

Objective: To study the structure of a two dimensional system with different interaction potentials.

Outline

1. Write a MC code for a (N,V,T) simulation of a Lennard-Jones (LJ) system in two dimensions. Use N = 256 atoms. Remember that the LJ potential is of the form: $U_{LJ}(r) = 4\varepsilon[(\sigma/r)^{12} - (\sigma/r)^6]$.
2. Create an initial set-up of positions from the sites of a square lattice.
3. Run a MC simulation for argon at T* =4.0 and ρ^*= 0.64. Compute the interaction energy U^*, the pressure P^*, and the radial distribution functions $g(r)$.
4. Repeat the simulation for the temperatures T*= 2.0 and T*=1.0, and compare with the previous results.
5. Repeat the same study for a system of soft disks, with an interaction potential given by

$$U_{sd}(r) = \begin{cases} U_{LJ}(r) + \varepsilon, & 0 < r \leq 2^{1/6}\sigma \\ 0, & r > 2^{1/6}\sigma \end{cases}$$

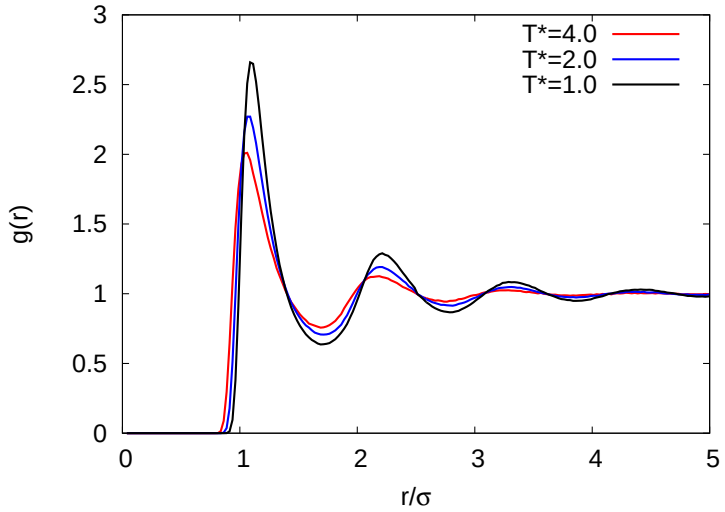
6. Run a MC simulation for a system of hard disks at the same density. In this case, the interaction potential is given given by

$$U_{hd}(r) = \begin{cases} \infty, & 0 < r \leq \sigma \\ 0, & r > \sigma \end{cases}$$

Lennard-Jones parameters

atom	mass(uma)	ϵ/k_B (K)	σ ($\mathring{A}$)
Ar	39.948	119.8	3.405

Some results to compare with:



Radial distribution functions for the Lennard-Jones system at different temperatures.

Questions

1. Give the expressions of the pressure P and the radial distribution $g(r)$ for the case of a two dimensional system.
2. Define the reduced units used in your simulations. Translate from reduced units to real units the values of T^*, ρ^*, U^* and P^*.
3. Discuss the definition given for the soft disks interaction potential.
4. Describe the implementation of the Metropolis algorithm for the hard disks system.